Model Selection and Evaluation Using Machine Learning Algorithms

1. Objective

The objective of this project is to compare multiple machine learning algorithms and determine the most suitable model for predicting passenger survival on the Titanic dataset. The models were evaluated using Accuracy, Precision, Recall, F1 Score, and Cross Validation Accuracy.

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2. Dataset Description

The Titanic dataset contains passenger information such as ticket class, gender, age, fare, and survival status.

Target Variable

- Survived

Features Used

- Pclass
- Sex
- Age
- SibSp
- Parch
- Fare
- Embarked

The dataset is a binary classification problem where the goal is to predict whether a passenger survived or not.

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3. Data Preprocessing

The following preprocessing steps were performed before training the models:

1. Missing values in the Age column were replaced using the median value.
2. Missing values in the Embarked column were replaced using the mode value.
3. Categorical variables were converted into numerical format using Label Encoding.

4. Relevant features were selected for model training.
5. The dataset was split into training data (80%) and testing data (20%).

These preprocessing steps ensured that the dataset was clean and suitable for machine learning algorithms.

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4. Models Implemented

Three machine learning models were trained and evaluated:

Logistic Regression

A statistical model commonly used for binary classification problems. It serves as a strong baseline model.

Decision Tree Classifier

A tree-based algorithm capable of learning non-linear relationships in data. However, it can be prone to overfitting.

Random Forest Classifier

An ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

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5. Evaluation Metrics

The following metrics were used to evaluate model performance:

Accuracy

Measures the proportion of correct predictions among all predictions.

Precision

Measures the percentage of predicted positive cases that were actually positive.

Recall

Measures the percentage of actual positive cases correctly identified.

F1 Score

The harmonic mean of Precision and Recall, providing a balanced measure of model performance.

Cross Validation Accuracy

Measures model performance across multiple folds and provides a more reliable estimate of generalization ability.

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6. Experimental Results

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**Cross Validation**
Logistic Regression	0.8101	0.7857	0.7432	0.7639	0.7890
Decision Tree	0.7821	0.7215	0.7703	0.7451	0.7801
Random Forest	0.8212	0.8088	0.7432	0.7746	0.8092

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7. Model Comparison

The results show that all three models achieved reasonable performance on the Titanic dataset.

Logistic Regression

- Accuracy: 81.01%
- F1 Score: 76.39%
- Cross Validation Accuracy: 78.90%

Logistic Regression provided a strong baseline and demonstrated stable performance.

Decision Tree

- Accuracy: 78.21%
- F1 Score: 74.51%
- Cross Validation Accuracy: 78.01%

Decision Tree captured non-linear relationships but produced lower overall performance compared to the other models.

Random Forest

- Accuracy: 82.12%
- F1 Score: 77.46%
- Cross Validation Accuracy: 80.92%

Random Forest achieved the highest Accuracy, F1 Score, and Cross Validation Accuracy among all evaluated models.

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8. Best Model Selection

The Random Forest Classifier was selected as the best-performing model because it achieved:

- Highest Accuracy (82.12%)
- Highest F1 Score (77.46%)
- Highest Cross Validation Accuracy (80.92%)

The ensemble nature of Random Forest allows it to combine the strengths of multiple decision trees while reducing overfitting, resulting in better generalization on unseen data.

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9. Visualizations Included

The following visualizations were generated and analyzed:

1. Confusion Matrix for Logistic Regression
2. Confusion Matrix for Decision Tree
3. Confusion Matrix for Random Forest
4. Model Performance Comparison Bar Chart

These visualizations helped in understanding prediction performance and comparing model effectiveness.

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10. Conclusion

This project successfully demonstrated the process of model selection and evaluation using machine learning techniques. Three classification algorithms were trained and compared on the Titanic dataset using multiple evaluation metrics and cross-validation.

Among the evaluated models, Random Forest achieved the best overall performance with an Accuracy of 82.12%, F1 Score of 77.46%, and Cross Validation Accuracy of 80.92%.

Therefore, Random Forest was selected as the final model for predicting passenger survival and is recommended for this classification task.